

A conserved distal-tail helical extension defines a tailspike attachment architecture in Gram-negative siphophages

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ABSTRACT

Rapid growth of bacteriophage genome collections has outpaced functional annotation of tail-tip proteins, limiting comparative analysis of host-recognition structures. Starting from a shared distal-tail gene organization in the *Salmonella* phages 9NA and Jersey, I developed a morphogenetic bioinformatic framework integrating gene synteny, sequence comparison, profile hidden Markov model (HMM) screening, structural evidence, structure-aware searching, and AlphaFold modeling. Comparison with the experimentally characterized lambda and Sf11 tail assemblies identified a predominantly alpha-helical C-terminal extension of the distal-tail (DT) protein associated with tailspike attachment, termed the distal-tail helical extension (DT-helix). Screening 541,986 proteins from 5167 complete NCBI RefSeq tailed-phage genomes, followed by evidence-based evaluation of sequence, genomic context, and structural architecture, identified 165 curated DT-helical-extension-associated phages. Their DT proteins segregated into six sequence groups. In the four principal multi-member groups, cognate tailspikes showed group-specific conservation in proximal N-terminal regions but substantially greater downstream diversity, consistent with sequence constraint at the DT–tailspike attachment boundary. A complementary ProST5/Foldseek search supported the established groups but revealed no convincing additional highly divergent family. Together with the experimentally characterized Sf11 attachment interface, these findings define a recurrent morphogenetic architecture linking conserved distal-tail scaffolds to more variable receptor-binding proteins across siphophages infecting Gram-negative bacteria. Although universal exchangeability is not established, the identified scaffold–receptor-binding boundaries provide a framework for molecular characterization and rational phage engineering. Accession-level information for the 165 curated phages is available through PhageTailDB.

1. Introduction

Salmonella enterica is a major bacterial pathogen responsible for illnesses ranging from self-limiting gastroenteritis to invasive systemic disease. Non-typhoidal *Salmonella* serovars are leading causes of food-borne infection, while invasive non-typhoidal disease disproportionately affects young children, immunocompromised individuals, and populations with limited access to healthcare (Stanaway et al., 2019). The emergence of resistance to fluoroquinolones, third-generation cephalosporins, and other frontline drugs has further reduced therapeutic options and has placed resistant *Salmonella* among bacterial pathogens requiring new antimicrobial strategies (World Health Organization, 2024).

Bacteriophages have consequently re-emerged as candidate precision antimicrobials (Kutateladze and Adamia, 2010; Nobrega et al., 2015). Their receptor-dependent specificity can restrict killing to

defined bacterial hosts, potentially reducing disruption of commensal microbiota, while replication in susceptible bacteria can amplify phage populations at sites of infection (Kortright et al., 2019). Clinical experience has established the feasibility of phage administration, although therapeutic use remains constrained by formulation, dosing, bacterial resistance, and the availability of phages active against individual isolates (Schooley et al., 2017; Dedrick et al., 2019; Jault et al., 2019; Leitner et al., 2021).

A central barrier to scalable phage therapy and synthetic phage design is the difficulty of predicting and modifying host specificity. In tailed phages, initial host recognition is mediated principally by receptor-binding proteins, including tail fibers and tailspikes, which engage surface structures such as lipopolysaccharides, capsules, outer-membrane proteins, and O-antigens (de Jonge et al., 2019). Tailspikes are especially important in many phages infecting enteric bacteria because they can combine receptor recognition with enzymatic

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degradation of surface polysaccharides. Altering receptor-binding proteins can redirect phage tropism, but a genetically precise replacement may still fail if the donor protein does not fold, oligomerize, attach to the virion, or function within the recipient tail architecture (Ando et al., 2015; Dunne et al., 2019; Yehl et al., 2019). The unresolved problem is therefore not only which protein recognizes a receptor, but which structural interface connects that protein to the conserved virion assembly.

This problem is compounded by inconsistent annotation of phage genes. Equivalent tail-associated proteins may be described as hypothetical proteins, generic tail proteins, distal tail proteins, tail fibers, or tailspikes, whereas similar labels may be applied to proteins with different morphogenetic roles. Sequence divergence further limits annotation pipelines that rely primarily on pairwise similarity. Conserved interfaces between tail assembly and receptor binding may therefore remain hidden even among complete phage genomes.

In this study, a morphogenetic approach was used to identify and characterize such an interface. The analysis began with a shared distal-tail gene organization in the *Salmonella* phages 9NA and Jersey and was interpreted by comparison with the experimentally characterized lambda tail (Wang et al., 2024) and Sf11 tailspike-attachment complex (Subramanian et al., 2025). A discovery-stage profile HMM was then used to screen 5167 complete NCBI RefSeq phage genomes, and candidate architectures were evaluated by integrating sequence support, genomic context, synteny, and structural features. The initially validated DT proteins were subsequently partitioned into six sequence groups, for which separate subgroup HMMs were constructed and used in a refined rescreening. This expanded analysis yielded a final curated set of 165 DT-helical-extension-associated phages. A related architecture had previously been visualized at lower resolution in the marine siphophage TW1, which carries six P22/Sf6-like tailspikes at its distal tail, although the molecular basis of spike attachment was not resolved (Wang et al., 2018). The more recent Sf11 structure provided local molecular detail for such an attachment architecture (Subramanian et al., 2025). Complementary ProstT5/Foldseek structure-aware searches were used to assess whether more highly divergent members had been missed, while AlphaFold Server modeling was used primarily to examine structural compatibility with the experimentally characterized Sf11 attachment architecture rather than as a classification criterion.

The resulting analysis identifies a conserved C-terminal helical extension of the distal-tail protein, termed the DT-helix, that marks a recurrent morphogenetic boundary between the distal-tail scaffold and associated tailspike proteins. Importantly, independently defined DT sequence groups are associated with corresponding sequence conservation in the proximal regions of their cognate tailspikes, whereas the downstream tailspike regions are substantially more diverse. I therefore evaluate the distribution, sequence diversification, and architectural variation of this tail-tip system as a morphogenetic class linking conserved tail assembly to variable receptor-binding appendages, and present PhageTailDB as a curated accession-level resource for comparative phage biology, structural studies, and future phage-engineering applications.

2. Materials and methods

2.1. Genome dataset

Complete genomes assigned to the class Caudoviricetes were retrieved from the NCBI RefSeq database in January 2026 using the query: “Caudoviricetes” AND “Annotated by NCBI RefSeq” AND “assembly level = complete”. The resulting accession-level dataset contained 5167 genomes (Table S1). Genome length, GC content, host metadata, and taxonomic assignments were extracted from the corresponding RefSeq records. Genome redundancy was assessed at two levels using Vclust (Zielezinski et al., 2025).

First, the complete set of 5167 RefSeq phage genomes was screened for exact nucleotide-sequence matches, treating reverse-complement sequences as equivalent. Five exact duplicate relationships were identified when the two genomes were reverse complements of each other. These records were flagged (Table S1, the last column) but retained as internal consistency controls, allowing DT-helix classification to be compared across independently named or accessioned records representing the same genome sequence.

Second, pairwise whole-genome similarities were subsequently calculated for the final set of 165 curated DT-helix-positive genomes, and complete-linkage clustering was performed at a threshold of 95% total average nucleotide identity (tANI). This analysis produced 159 clusters, including 154 singletons and five multi-member clusters: one three-genome cluster and four two-genome clusters (Table S3, Section A, the last column). A strictly dereplicated dataset would therefore contain 159 entries, representing a reduction of only six genomes (3.64% of the curated collection). All records were retained in PhageTailDB because identical or highly similar genomes may appear in the literature and public databases under different phage names or accession identifiers.

2.2. Sequence comparison and morphogenetic annotation

Protein sequences and gene neighborhoods were obtained from the RefSeq records. Candidate relationships were evaluated using BLASTP by BLAST 2.16.0+ with default parameters (Altschul et al., 1990, 1997; Camacho et al., 2009). Multiple-sequence alignments were generated with MUSCLE version 3.8.31 (<http://www.drive5.com/muscle/>) with default parameters. Functional interpretation did not rely on deposited protein names alone. Instead, proteins were assigned to morphogenetic categories by integrating sequence similarity, gene order relative to the tape-measure and receptor-binding genes, predicted domain composition, structural similarity, and published experimental evidence.

Distal-tail gene order was compared across the 165 validated genomes after normalizing loci to the morphogenetic direction from the tape-measure gene downstream. Recurring arrangements were summarized from the assigned TM, DT, TH, TNLP, CF, and tailspike components, with absent or unannotated components recorded without affecting the relative order of the remaining genes.

2.3. Profile-HMM construction, genome screening, and clustering

First, a seed alignment was created from the multiple sequence alignment of DT proteins of 9NA, Jersey, FSL SP-031, K1H, K1ind1, and Sf11. The alignment was truncated to the box region that included the conserved DT core and a helical extension shown in Fig. 1a. A profile HMM was constructed with HMMER 3.3.2 using hmmbuild (Eddy, 2011). Proteins from all 5167 genomes were searched with hmmsearch using a reporting threshold of E-value ≤ 10.0 . The raw output, seed alignment, and final HMM profile are available at [<https://github.com/tlensk/Sal3T>, in 1_HMM_initial/]. The HMM hits were manually curated to yield a set of 160 validated DT-helix proteins.

Second, I clustered the 160 validated DT-helix proteins using MMseqs2 (minimum sequence identity = 0.2, minimum alignment coverage = 0.2, clustering mode = “normal- > slow_sensitive”) (Steinberger and Söding, 2017). Then the clustering results of the validated DT-helix proteins were employed for refining the HMM search. For each non-singleton DT-helix cluster, a separate seed alignment was created from a reduced set of representative sequences generated by MMseqs2 (minimum sequence identity = 0.8, minimum alignment coverage = 0.8, clustering mode = “normal- > slow_sensitive”); for two singleton DT-helix clusters, the corresponding HMMs were built using one sequence; Fig. 1b shows the MSA/sequences and AF models of the representative sequences for each cluster. Then, the proteins from all 5167 genomes were screened again using this set of six HMMs. The clustering results, raw output, seed alignments, and final HMM profiles are available at [<https://github.com/tlensk/Sal3T>, in

(a)



(b)

Cluster A

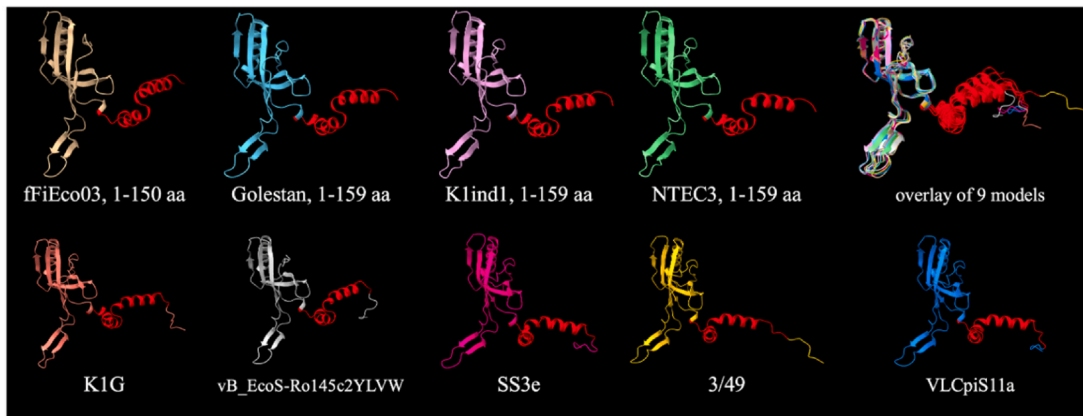
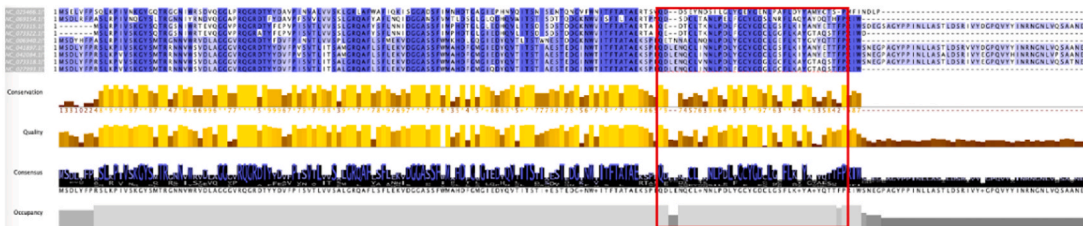


Fig. 1. (a) MSA of DT sequences from three Salmonella phages: 9NA, Jersey, and FSL SP-031, two Escherichia phages: K1H and K1ind1, and Shigella phage Sf11; the dark grey rectangle indicated the part of the alignment used for creating the initial HMM; the light blue area represents the almost identical sequences shared by K1ind1 and K1H phages in the DT core and helical extension region, the key difference is the C-term appendage of K1ind1 that is missing in K1H. The light grey area represents the region of unknown function shared between K1ind1 and Sf11;

(b) for clusters A-F, MSAs and AF models are shown; for clusters E-F, single sequence and AF model are shown; an approximate location of helical extensions is highlighted in red color and by red boxes; MSAs and sequences are visualized in Jalview 2.11.5.2 (Waterhouse et al., 2009). Panel (b) shows the six groups defined from the initial 160-protein set and used to construct the subgroup HMMs. Final group assignments after rescreening are provided in Table S6 (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

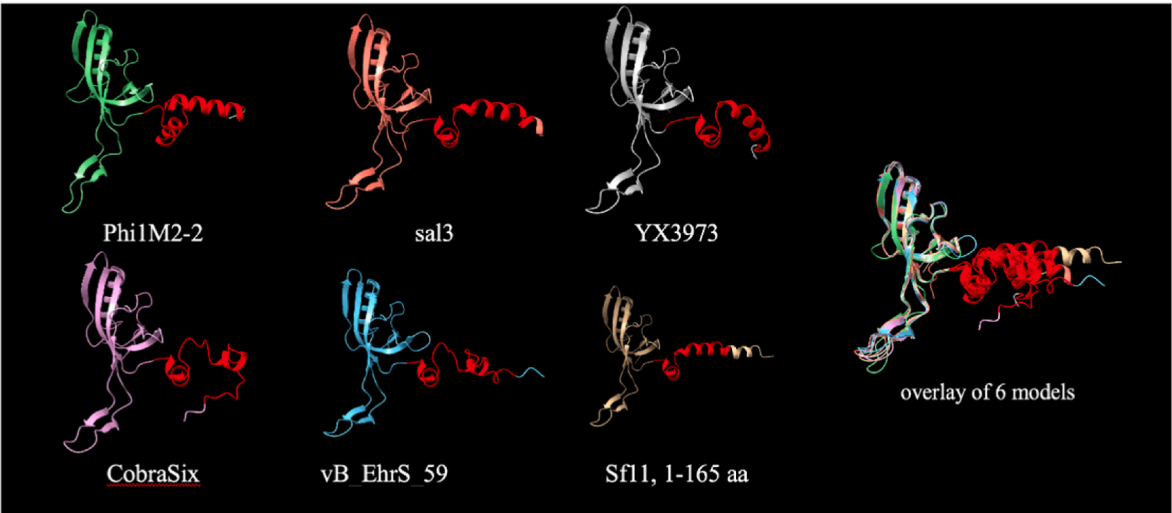
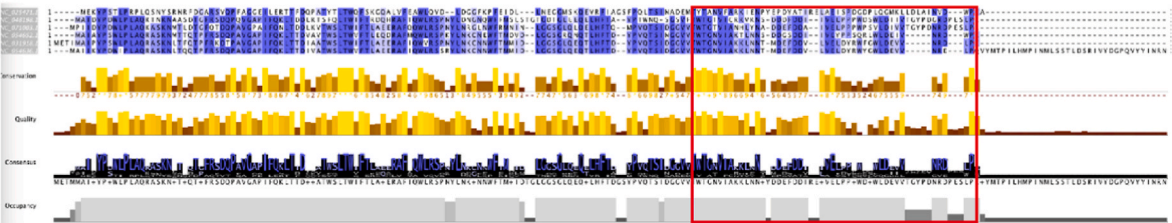
2_HMM_re-screen].

2.4. Structure-aware search using a local ProST5/Foldseek database

To complement the profile-HMM searches and assess whether highly sequence-divergent proteins with a DT-helical-extension-like structural signature might have been missed, a complementary structure-aware

screening was performed using Foldseek (van Kempen et al., 2024) locally (commit 7f4b94644d6b210af4530e04a9fdc9e4bb64ea66) with an E-value threshold of $1e-3$ and $-\text{max-seqs}$ 1000. A local Foldseek ProST5 database was constructed from all 541,986 proteins encoded by the 5167 complete phage genomes included in the dataset. Protein sequences were converted to Foldseek-compatible 3Di structural-alphabet representations using ProST5, allowing structure-aware comparison

Cluster B



Cluster C

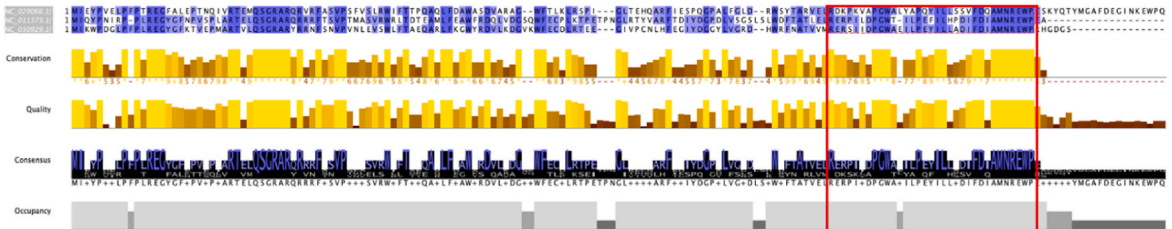
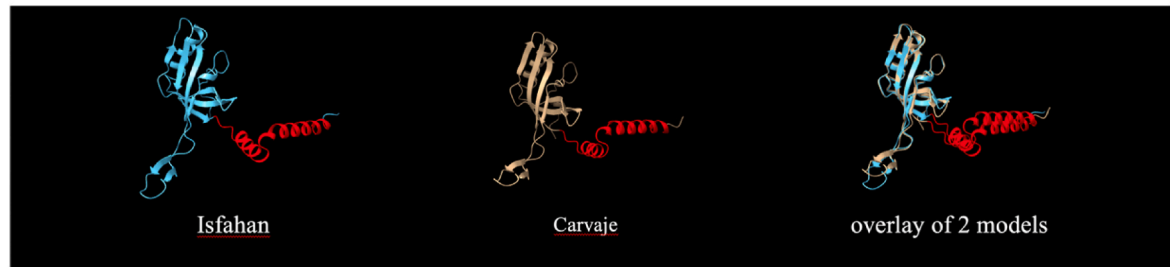
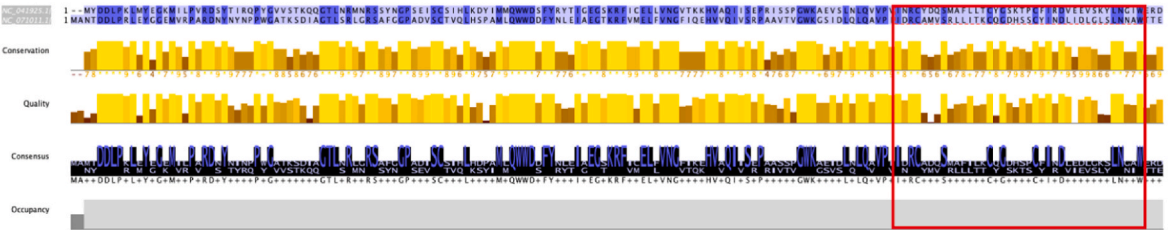


Fig. 1. (continued).

without generating full three-dimensional coordinate models for every protein. AF models of 22 representative DT proteins (Fig. 1b) from the six clusters were used as queries against this local database. Because ProST5 infers the 3Di representation from amino-acid sequence, this analysis was treated as complementary structure-aware evidence rather than as a fully sequence-independent coordinate-based structural

search. Information about the procedure and raw output files related to the structure-aware screening are available at [<https://github.com/tlensk/Sal3T>, in 3_structure-aware_screen].

Cluster D



Cluster E

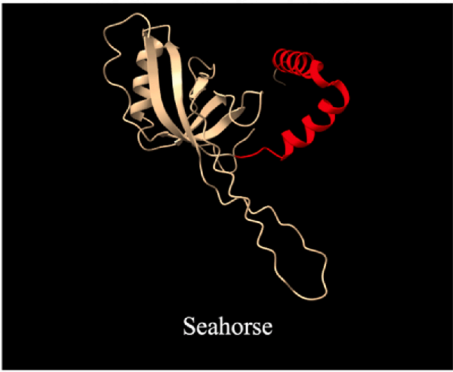


Fig. 1. (continued).

2.5. Structure prediction and comparison

Protein monomers and DT-tailspike complexes were modeled with AlphaFold Server (AF) [Release 2026.04.16, <https://alphafoldserver.com/>] (Abramson et al., 2024). Multimer predictions were performed with a stoichiometry corresponding to the biologically inferred distal tail and tailspike assembly (DT monomer and tailspike trimer). Model quality was assessed using predicted TM-score (pTM), interface predicted TM-score (ipTM), per-residue confidence (pLDDT), predicted aligned error (PAE), and interaction prediction score from aligned errors

(ipSAE) (Dunbrack, 2025). The AF models for the predicted multimer DT-tailspike complexes for the initial six representative phages are shown in Table 3 and Fig. S1. Protein MSAs generated by AlphaFold Server are derived from searches of UniRef90, MGnify, and small-BFD. Exact release identifiers for the database snapshots used by the public server were not available in the downloaded prediction metadata and therefore could not be reported. Experimental structures of lambda and Sf11 used for benchmarking were obtained from the Protein Data Bank [PDB IDs: 8IYK; 9OCC]. Predicted and experimental structures were superimposed with UCSF

Cluster F

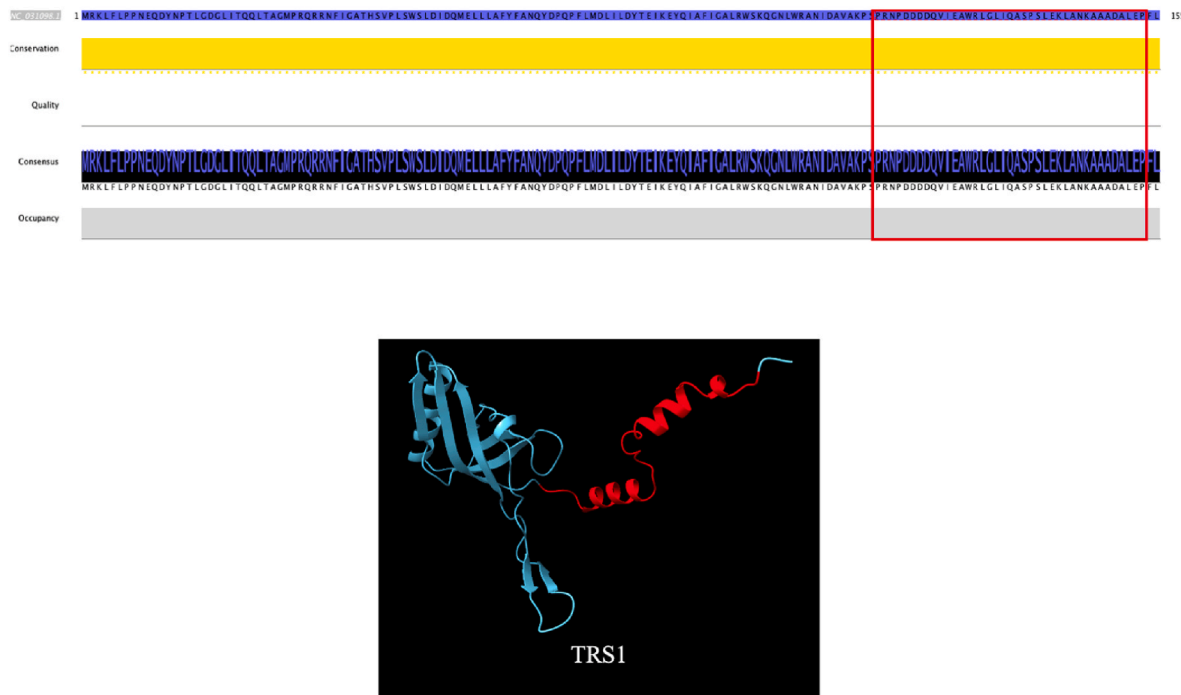


Fig. 1. (continued).

ChimeraX v. 1.13 MatchMaker [<https://www.rbvi.ucsf.edu/chimeraX>], and α RMSDs were reported before and after iterative pruning. AlphaFold confidence metrics were used descriptively rather than as binary interaction criteria. In particular, global ipTM was not used to assess the DT–tailspike interface because it incorporates all inter-chain relationships within the complex, including the extensive interfaces of the trimeric tailspike. Pairwise DT–tailspike confidence was examined where available but was likewise not used as an acceptance threshold. Complex predictions were instead evaluated for recurrence and broad geometric compatibility with the experimentally characterized Sf11 attachment architecture. For comparison of selected Sf11 interface contacts, residue pairs reported in the 9OCC attachment interface were evaluated in the experimental structure and across five AF models. Distances were calculated between the closest chemically equivalent charged side-chain atoms for each residue pair (Glu OE1/OE2, Lys NZ, and the relevant Arg guanidinium nitrogen atoms).

In four cases, *Acinetobacter* phage vB_AbaS_TRS1 (NC_031098.1), *Idiomarinaceae* phage Phi1M2-2 (NC_025471.1), *Pseudoalteromonas* phage TW1 (NC_070822.1), and *Janthinobacterium* phage vB_JliS-Donnerlittchen (NC_073031.1), AF models could not recover the biologically relevant placement of DT monomer within the DT–tailspike complex. For TW1, this failure is particularly informative because published cryo-EM data independently demonstrate six trimeric tailspikes attached to the distal tail complex (Wang et al., 2018). Thus, failure to recover a plausible AF complex geometry was not treated as evidence against DT–helical-extension classification. Consequently, PhageTailDB shows the predicted DT structure and tailspike trimer structure for these four phages.

To assess whether an independent molecular-docking approach could reproduce the experimentally characterized DT–tailspike attachment geometry, de novo protein–protein docking was performed using the HADDOCK 2.4 Web Server (Honorato et al., 2024). The *Shigella* phage Sf11 gp17–gp13 complex was used as a positive control because

its attachment interface is experimentally defined in the cryo-EM structure PDB 9OCC. The resolved proximal region of gp17 (chain C, residues 3–165) and the N-terminal attachment region of the gp13 tailspike trimer (chains G–I, residues 4–59) were extracted and submitted as separate docking partners. Docking was performed without user-defined residue-contact restraints so that recovery of the known interface could serve as an independent test of the docking procedure rather than being imposed by prior structural information.

Generated docking solutions were examined for placement of the gp17 C-terminal helical-extension region relative to the N termini of the gp13 trimer and for recovery of the experimentally observed interface geometry. Particular attention was given to the characteristic helical packing at the gp17–gp13 attachment site. Because the de novo docking procedure did not reliably reproduce the known Sf11 interface, HADDOCK scores and estimated interaction energetics were not used to evaluate the uncharacterized DT–tailspike systems. Restrained docking based on contacts derived from the Sf11 cryo-EM structure or AlphaFold models was not pursued because such restraints would incorporate the interface geometry being tested and therefore would not constitute independent validation. Docking inputs and outputs are available at [<https://github.com/tlensk/Sal3T>, in docking].

2.6. Evidence-based classification of candidate DT–helical-extension proteins

Candidates recovered by profile-HMM and structure-aware searches were evaluated using three complementary classes of evidence: protein-sequence evidence, genomic-context evidence, and structural evidence. Each evidence class was scored as supporting (1), indeterminate (0), or contradictory (–1) with respect to DT–helical-extension classification. **Protein-sequence evidence** included HMM profile support, alignment extent and position, and protein length relative to the validated DT–helix proteins. Particular attention was given to whether the profile match

spanned a substantial portion of the DT core with extension region rather than representing a short or compositionally biased local match. Compact DT proteins and longer proteins containing additional C-terminal appendages were evaluated separately rather than subjected to a single universal protein-length cutoff. **Genomic-context evidence** evaluated the position of the candidate within the tail morphogenesis locus, including its relationship to the tape-measure gene, neighboring tail-tip proteins, recurrent local gene-synteny patterns, and the presence and position of a candidate receptor-binding protein or tailspike gene. Because tail-tip gene complements differ among phage lineages, correspondence to the overall morphogenetic organization rather than identity to a single gene-order pattern was required. **Structural evidence** evaluated whether the candidate was compatible with the conserved DT fold and contained a C-terminal region of predominantly alpha-helical character extending beyond the DT core. The internal number, boundaries, and relative lengths of the helices were not required to be identical among sequence groups. Predicted trimerization of associated tailspikes and recurrent DT–tailspike complex geometries were recorded as supporting observations but were not used as independent criteria for physical interaction or for classification. Global ipTM, pairwise ipTM, and ipSAE values were not used as pass/fail thresholds.

Candidates supported by all three evidence classes were classified as confirmed DT-helical-extension architectures. Candidates with mixed evidence were designated borderline when the available evidence remained compatible with a DT-related architecture but one or more evidence classes were indeterminate or discordant. This category included candidates supported by two evidence classes with incomplete evidence in the third, as well as candidates with supportive genomic context, indeterminate sequence evidence, and atypical structural features warranting further investigation. Candidates showing clear incompatibility across two or more evidence classes were rejected.

2.7. Pairwise sequence-identity analysis of DT proteins and associated tailspikes

Pairwise amino-acid sequence identities were calculated for the 165 curated DT proteins and their corresponding tailspikes using custom scripts. Three sequence datasets were analyzed separately: full-length DT proteins, the N-terminal 60 residues of each associated tailspike, and the remaining downstream tailspike sequence (residues 61–end). In genomes lacking an annotated tailspike CDS, the corresponding sequences were rescued manually based on their expected position within the conserved tail-tip genomic neighborhood. Known tailspike sequences were used as templates to align against the surrounding genome sequence, probing all six reading frames to identify missing or unannotated portions of the CDS and reconstruct the corresponding tailspike. Cognate tailspikes were identified primarily by their position within the conserved tail-tip genomic neighborhood, with deposited annotation, sequence homology, and predicted structure used as supporting evidence. Tailspikes were assigned to groups A–F according to the sequence-group assignment of their cognate DT proteins. For each dataset, every pair of sequences was compared by global pairwise alignment. Percent identity was calculated as the percentage of identical residue-to-residue aligned positions, excluding positions containing a gap in either sequence from the denominator. Self-comparisons were assigned 100% identity. The resulting symmetric pairwise-identity matrices therefore contained identity values for all sequence pairs. For group-level comparisons, self-comparisons were excluded. Within-group distributions consisted of all unique pairwise comparisons between sequences assigned to the same DT group, whereas between-group distributions consisted of comparisons between sequences assigned to different DT groups. Median pairwise identities were calculated for each group and sequence region. To quantify the relative conservation of the proximal tailspike region, the N-terminal and downstream identities were also compared for the same tailspike pairs, and the proportion of

within-group pairs for which N-terminal identity exceeded downstream identity was calculated. Singleton group was not evaluated for within-group pairwise conservation.

Pairwise-identity matrices were visualized as heatmaps without hierarchical clustering, seriation, or other automatic reordering. Rows and columns retained the sequence order of the input FASTA files, which was identical across the DT, N-terminal tailspike, and downstream tailspike datasets, allowing direct visual comparison among the three matrices (Table S6). Higher percent-identity values were represented by greater color intensity. The labeled pairwise-identity matrices and corresponding heatmaps were exported as CSV and PNG files, respectively, and are available at [<https://github.com/tlensk/Sal3T>, in *tailspike_analysis*].

2.8. Genome visualization and PhageTailDB

Local genome maps were generated with vGV version 1.0.1 [<https://tlensk.github.io/vgv/>;

DOI: 10.5281/zenodo.21133263] and standardized to a common orientation for comparison. PhageTailDB was developed to provide supporting information about DT-helix classification. The resource contains genome metadata, host information, and visualization of the predicted DT and tailspike structures for 165 curated phages. The PhageTailDB v8.2 is accessible at [<https://tlensk.github.io/phagetaildb-dev/>; DOI: 10.5281/zenodo.21921997].

3. Results

3.1. Similar distal-tail regions in phages 9NA and Jersey reveal a shared morphogenetic organization

My comparative inspection of published *Salmonella* phage genomes identified a similar distal-tail gene arrangement in phage 9NA (NC_025443.1) and phage Jersey (NC_021777.1). These phages represent independently characterized groups: the 9NA-like (Zeng et al., 2019) phages and the former *Jerseyvirinae* (Anany et al., 2015), now *Guernseyvirinae*. In both genomes, tape-measure protein coding gene was followed by three comparatively small open reading frames and a larger tail-associated gene with reported remote similarity to the lambda gene coding gpJ. A tailspike gene was located immediately downstream of this cluster, although the tailspike was encoded on opposite strands in these two genomes (Fig. 2).

The original GenBank records did not assign equivalent functional labels to all proteins occupying corresponding positions (Fig. 2). Several were annotated only as hypothetical or generic tail proteins, preventing recognition at the level of deposited names. However, conservation of gene order, protein length, and location between the tape-measure and tailspike genes suggested that these open reading frames represented a shared distal-tail morphogenetic module.

3.2. Comparison with lambda helps assign morphogenetic roles and identify a distinct C-term helical extension

The distal tail of phage lambda provides an experimentally characterized reference composed of Distal Tail (DT) gpM, Tail Hub (TH) gpL, tail NlpC-like protein gpK (TNLP), Tail Hub Insert (THI), and gpJ. Tail-tip terminology requires particular care in a comparative morphogenetic study because homologous morphogenetic domains may be distributed among different proteins in different phages. In lambda, gpL and the proximal portion of gpJ together contribute to the baseplate/tail-tip hub, whereas additional C-terminal gpJ domains form the central fiber and receptor-binding region. Therefore, proteins homologous to the hub-forming region of lambda gpJ are termed here a gpJ-like hub protein contributing to the central funnel-shaped tail-tip structure (CF); and the term ‘central fiber’ is reserved for a structurally or functionally supported central fiber structure or domain. Consistent with this interpretation, a recent comparative analysis of the siphophage LSH1

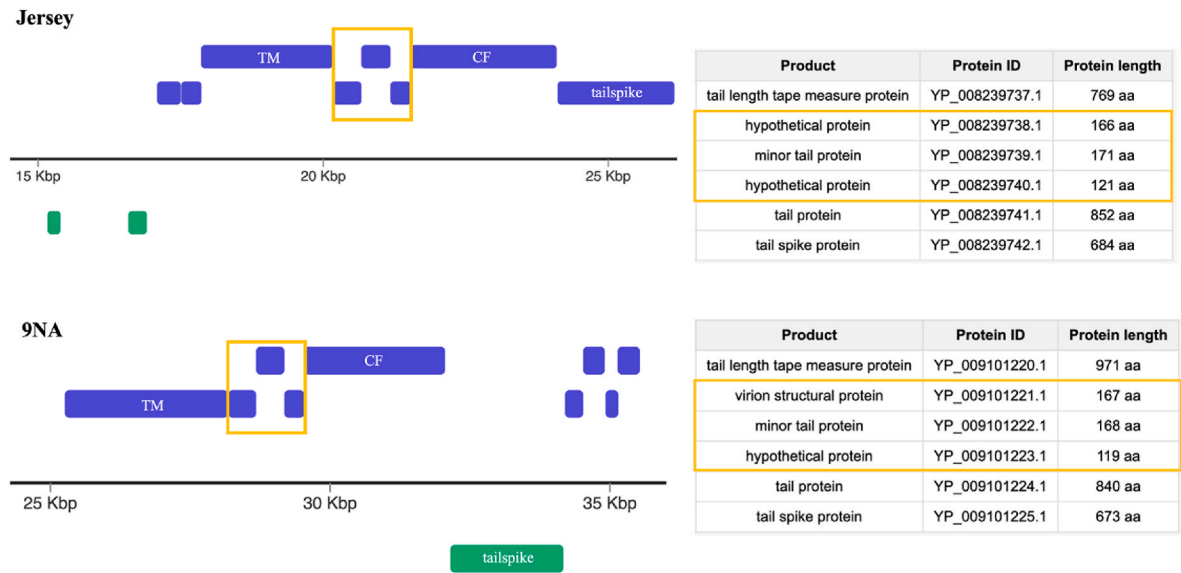


Fig. 2. Genome maps of the distal tail cluster in the Salmonella phages, 9NA and Jersey (genomic maps are generated using vGV v.1.0.1). Gene orientation is indicated by color: ‘+’ strand in lilac; ‘-’ strand in mint. Tables beside the genome maps contain GenBank annotation information about the genes discussed above that are highlighted in amber. The order of the genes in the table follows the order of genes in the corresponding genome GenBank file. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

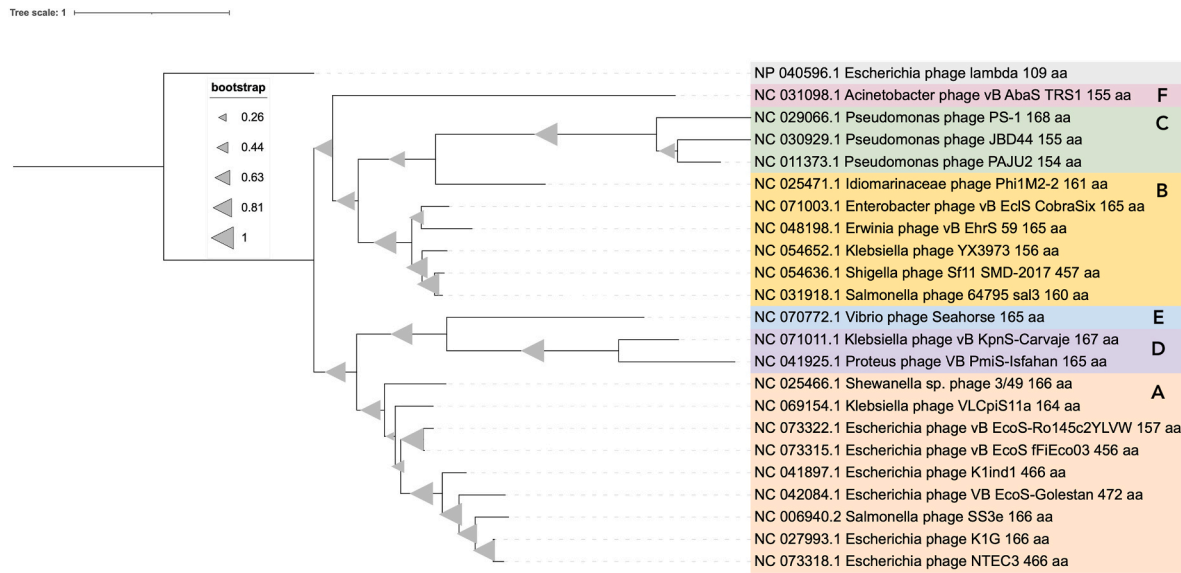


Fig. 3. The sequence tree of 22 representative DT-helix proteins. The tree is rooted using DT protein of phage lambda as an outgroup and visualized using iTOL.

assigned Jersey gp30 and gp32 as gpL-like and gpJ-like hub components, respectively, and indicated that the Jersey gpJ-like protein lacks the distal antireceptor extension characteristic of the longer lambda-like gpJ architecture (Hardies et al., 2026). To evaluate whether predicted structures could recover tail-tip components, I compared AF models of the lambda proteins with the published cryo-EM structures (Wang et al., 2024). Predicted and experimental protein structures were superimposed using UCSF ChimeraX v1.13 (Meng et al., 2023). Structural agreement was reported as the RMSD of α pairs retained after iterative pruning and the RMSD across all sequence-aligned α pairs.

Table 1 shows that AF reproduced the experimentally resolved core folds of the lambda distal-tail proteins with variable accuracy. The gpM prediction showed strong agreement with the cryo-EM structure, with a core α RMSD of 0.632 Å over 84 pruned residue pairs and an all-pairs RMSD of 2.932 Å over 109 residues. The gpL model also closely

reproduced the central tail-hub core, with a core RMSD of 0.832 Å over 142 residue pairs, although the much higher all-pairs RMSD of 16.306 Å indicated substantial differences in peripheral or flexible regions. The full-length gpK model had high predicted confidence (pTM = 0.90), but no direct structural comparison was possible because gpK was not resolved in the cryo-EM assembly. In contrast, the extended Tail Hub Insert gpI was predicted with low confidence (pTM = 0.41), consistent with its flexible and assembly-dependent conformation.

Using the lambda proteins as morphogenetic references, the three small proteins in 9NA and Jersey were assigned to the distal tail (DT), tail hub (TH), and TNLP categories (Table 2). The corresponding proteins retained the expected order and expected core folds but differed from lambda in several reproducible features (Table S5). Most notably, the 9NA and Jersey DT proteins contained a C-terminal helical extension that was not present in lambda gpM. Their TH proteins lacked the

Table 1
Structural benchmarking and confidence assessment of lambda distal-tail proteins. The AF models are colored by per residue pLDDT confidence score using ChimeraX.

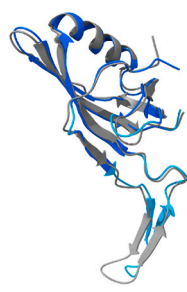
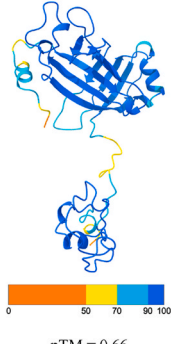
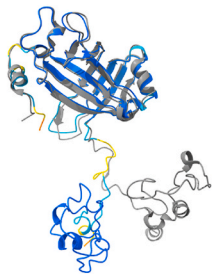
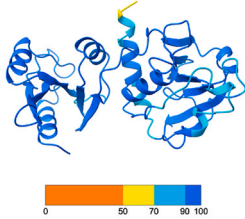
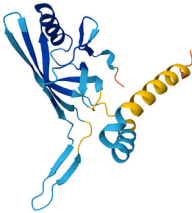
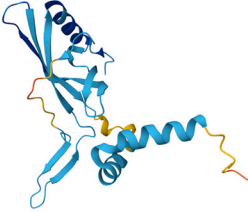
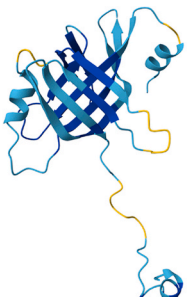
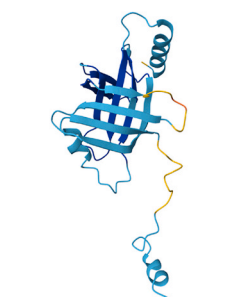
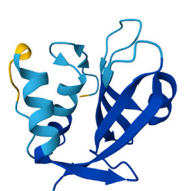
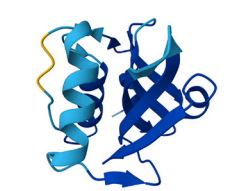
Protein	cryoEM (Wang et al., 2024)	AF (current study)	Comparison
gpM – Distal Tail Length: 109 aa	 PDB: 8IYK, chain M	 pTM = 0.82	 AF (per residue pLDDT confidence score coloring) vs. cryoEM (grey): core Cα RMSD, 0.632 Å over 84 pruned residue pairs; all-pairs RMSD, 2.932 Å over 109 aligned residue pairs.
gpL – Tail Hub Length: 232 aa	 PDB: 8IYK, chain L	 pTM = 0.66	 AF(blue) vs. cryoEM (orange): core Cα RMSD, 0.832 Å over 142 pruned residue pairs; all-pairs RMSD, 16.306 Å over 232 aligned residue pairs.
gpK – TNLP Length: 247 aa	N/A gpK was not resolved in the lambda tail cryo-EM structure	 pTM = 0.9 The full-length lambda gpK sequence referenced by Coburn (2011) was used for AF prediction.	N/A
gpI – Tail Hub Insert Length: 223 aa	 PDB: 8IYK, chain I	 pTM = 0.41	Prediction of the small THI protein produced very low overall structural agreement (pTM-score 0.41), consistent with its extended and context-dependent conformation.

Table 2
The AF models of the distal-tail proteins for 9NA and Jersey. The models are colored by per residue pLDDT by AlphaFold Server.

Protein category	9NA		Jersey	
Distal Tail (DT)		Protein ID: gp51 [YP_009101221.1] Length: 167 aa pTM = 0.68		Protein ID: gp29 [YP_008239738.1] Length: 166 aa pTM = 0.67
Tail Hub (TH)		Protein ID: gp52 [YP_009101222.1] Length: 168 aa pTM = 0.74		Protein ID: gp30 [YP_008239739.1] Length: 171 aa pTM = 0.72
TNLP		Protein ID: gp53 [YP_009101223.1] Length: 119 aa pTM = 0.82		Protein ID: gp31 [YP_008239740.1] Length: 121 aa pTM = 0.86

Very high (pLDDT > 90)

Confident (90 > pLDDT > 70)

Low (70 > pLDDT > 50)

Very low (pLDDT < 50)

lambda C-terminal region associated with the 4Fe-4S cluster, and their TNLP proteins corresponded primarily to the C-terminal domain of the two-domain lambda gpK. Neither Salmonella phage contained an obvious counterpart of the lambda THI protein as a separate protein. In 9NA and Jersey, there is a tailspike gene immediately downstream of CF. These differences indicate that 9NA and Jersey share a distal-tail architecture distinct from, but morphogenetically comparable to, the lambda system.

3.3. Related *Guernseyvirinae* and *Sf11* connect the DT helical extension to tailspike attachment

I next examined additional representatives related to phage Jersey. Salmonella phage FSL SP-031 (NC_021775.1), a representative of *Cornellvirus*, and Escherichia phages K1H (NC_027994.1) and K1ind1 (NC_041897.1), representatives of *Kagunavirus*, contained the same TM-DT-TH-TNLP-CF organization followed by a candidate tailspike gene. The DT proteins of FSL SP-031 and K1H were nearly identical to the Jersey DT protein apart from a small number of substitutions (Fig. 1a). K1ind1 retained the DT core and helical extension but also encoded a relatively large C-terminal appendage, producing a DT protein approximately three times longer than those of 9NA, Jersey, FSL SP-031, and K1H.

The extended K1ind1 protein resembled gp17 of Shigella phage Sf11 in sequence distribution and predicted domain organization. The cryo-EM structure of Sf11 tail has shown that gp17 forms part of the distal tail assembly and uses an adaptor region to attach the tailspike trimer

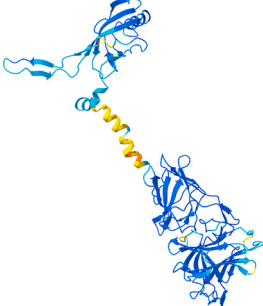
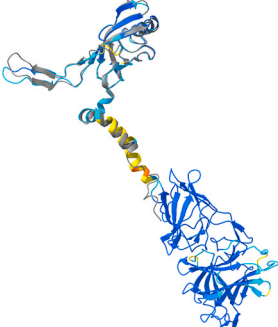
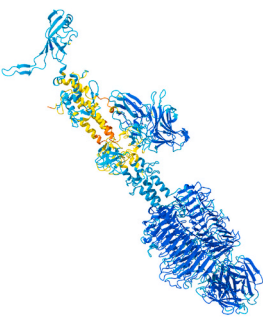
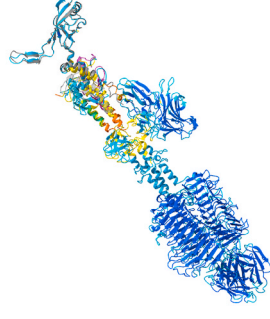
(Subramanian et al., 2025). Alignment of the representative DT proteins localized the shared DT core and helical extension, while the longer K1ind1 and Sf11 proteins contained an additional C-terminal appendage of uncertain function (Fig. 1a). Throughout this study, “C-terminal helical extension” and “helical extension” refer to the helical region immediately following the conserved DT core. In longer DT proteins such as Sf11 gp17 and K1ind1 gp26, additional sequence resulting in the C-terminal appendage extends beyond this region.

3.4. AF predicts analogous DT-extension-tailspike interfaces in representative phages

The Sf11 complex was first used as a structural benchmark. The AF model closely reproduced the experimentally resolved region of Sf11 gp17 containing the helical extension (Table 3, top row). Superposition with gp17 residues 3–165 of chain C in the 90CC cryo-EM structure yielded a core Cα RMSD of 0.818 Å over 137 pruned residue pairs and an RMSD of 2.584 Å across all 163 aligned pairs. These values demonstrate accurate recovery of the experimentally resolved gp17 fold, whereas the remaining C-terminal portion of the 457-residue protein could not be assessed because it was not resolved in the cryo-EM model.

To evaluate the predicted relative placement of the tailspike independently of the gp13 trimer itself, the AF gp17 chain was first superimposed onto experimental gp17 residues 3–165, and the same transformation was then applied to the complete predicted gp17–gp13 complex (Table 3, the bottom row). Without further fitting of the tailspike chains, the experimentally resolved N-terminal regions of the three

Table 3
The comparison of AF models with the published cryo-EM tail structure (Subramanian et al., 2025) of Sf11.

Protein	cryoEM (Subramanian et al., 2025)	AF(current study)	Comparison
gp17 – Distal Tail Length: 457 aa	 [dark grey] PDB: 9OCC, chain C, residues 3-165	 [per residue pLDDT confidence score coloring] pTM = 0.65	 AF vs. cryoEM: core Cα RMSD, 0.818 Å over 137 pruned residue pairs; all-pairs RMSD, 2.584 Å over 163 aligned residue pairs.
Distal Tail - gp17 Length: 457 aa <i>cryoEM:</i> <i>residues 3-165</i> + tailspike - gp13 Length: 585 aa <i>cryoEM:</i> <i>residues 4-59</i>	 PDB: 9OCC gp17 – chain C [dark grey] (residues 3-165) gp13 trimer – chains G,H,I [teal, magenta, grey; 50% transparent] (residues 4-59)	 gp17, full length gp13 trimer, full length [per residue pLDDT confidence score coloring] ipTM = 0.59, pTM = 0.64	 Superposition of the full-length AF model (gp17 monomer and gp13 trimer) onto gp17 residues 3–165 of the cryo-EM structure: core Cα RMSD, 0.647 Å over 128 pruned residue pairs; all-pairs RMSD, 3.044 Å over 163 aligned residue pairs.

gp13 subunits (residues 4–53) were positioned with a combined Cα RMSD of 5.85 Å over 150 residue pairs. Thus, AF recovered the overall location and orientation of the proximal gp13 trimer relative to gp17, but substantially less accurately than it recovered the gp17 fold itself. The Sf11 benchmark therefore supports broad geometric compatibility of the predicted DT–tailspike arrangement, rather than atomic accuracy

of the predicted interface.

This distinction was also apparent at the residue level. Four gp17–gp13 contacts identified in the experimental 9OCC model were examined across five AF models using the minimum distance between chemically equivalent charged side-chain atoms. In the cryo-EM model, the corresponding distances were 5.20 Å for gp17 E129–gp13 R16,

Table 4
Comparison of experimentally modeled Sf11 gp17–gp13 contacts with AlphaFold predictions.

Sf11 interface contact	9OCC experimental distance (Å)	AF model 0 (Å)	AF model 1 (Å)	AF model 2 (Å)	AF model 3 (Å)	AF model 4 (Å)	AF mean ± SD (Å)
gp17 E129 – gp13 R16	5.197	9.241	7.554	8.823	8.446	6.872	8.19 ± 0.96
gp17 E136 – gp13 K19	4.279	5.436	6.302	6.860	7.752	4.566	6.18 ± 1.24
gp17 E147 – gp13 K19	2.254	2.427	2.107	2.295	1.940	2.298	2.21 ± 0.19
gp17 E152 – gp13 K19	2.730	4.629	4.483	4.942	4.374	4.990	4.68 ± 0.27

^a Distances were measured between the closest chemically equivalent charged side-chain atoms for each residue pair. For Glu (E), OE1/OE2 atoms were considered; for Lys (K), NZ; and for Arg (R), the relevant guanidinium nitrogen atoms. The experimental values were measured in the locally modeled Sf11 gp17–gp13 interface of PDB 9OCC. AlphaFold values were measured independently in five AlphaFold Server models of the gp17 monomer–gp13 trimer complex. Mean values and standard deviations were calculated across the five models.

^b Experimental residue pairs correspond to gp17 chain C and gp13 chains G–I as follows: E129–R16, chain H; E136–K19, chain G; E147–K19, chain I; and E152–K19, chain I.

4.28 Å for E136–K19, 2.25 Å for E147–K19, and 2.73 Å for E152–K19 (Table 4). Across the five AF models, the corresponding mean distances were 8.19 ± 0.96 Å, 6.18 ± 1.24 Å, 2.21 ± 0.19 Å, and 4.68 ± 0.27 Å, respectively. Thus, the E147–K19 interaction was recovered closely and reproducibly, whereas E136–K19 was only partially reproduced and the E129–R16 and E152–K19 contacts were substantially looser than in the experimental structure. These comparisons indicate that AF captures the broad topology and part of the local interaction network but does not reproduce the complete Sf11 interface at atomic accuracy.

Consistent with this interpretation, interface-specific confidence remained low despite the experimentally established interaction. Across the five Sf11 models, pairwise gp17–gp13 ipTM values were approximately 0.17–0.21, and principal ipSAE values at a 10 Å PAE cutoff were approximately 0–0.013. These metrics were therefore treated as confidence diagnostics rather than as evidence for or against physical interaction. Together, the direct structural comparison and confidence analysis support use of AF complexes to examine plausible relative geometry in homologous systems, but not to validate binding or infer specific residue-level contacts.

To assess whether an independent molecular-docking approach could provide additional support for the DT–tailspike interaction, I used HADDOCK 2.4 Web Server (Honorato et al., 2024) as a representative integrative protein–protein docking framework in a restraint-free test. The experimentally characterized Sf11 system served as a positive control to determine whether the known attachment geometry could be recovered without supplying interface-specific restraints. The docking used the resolved/proximal regions of gp17 and the N-terminal gp13 trimer corresponding to the attachment region. Although several solutions positioned the gp17 helical-extension region near the N termini of the gp13 trimer, the docking did not reliably reproduce the experimentally observed Sf11 attachment geometry, including the characteristic four-helix-bundle packing at the gp17–gp13 interface. Because HADDOCK docking was not successful in recovering the known interface of this experimentally supported positive control, docking scores and estimated interaction energetics were not considered reliable criteria for evaluating the uncharacterized DT–tailspike systems. Restrained docking using contacts derived from the Sf11 structure or AlphaFold models was not pursued because such restraints would incorporate the interface geometry being tested and therefore would not provide independent validation. More information is available at [https://github.com/tlensk/Sal3T, in docking].

3.5. Profile-HMM screens identify 165 curated DT-helix-associated architectures

To determine the prevalence of the architecture, I constructed a profile HMM from a curated alignment of six representative DT proteins: 9NA, Jersey, FSL SP-031, K1H, K1ind1, and Sf11 (Fig. 1). The initial seed alignment was derived from the Multiple Sequence Alignment by truncating it to the conserved DT core with helical-extension region to avoid over-weighting the appendages present in K1ind1 and Sf11. The profile was used to search 541,986 proteins from 5167 complete Caudoviricetes genomes available in NCBI RefSeq in January 2026 (Table S1).

The initial search returned 166 proteins with at least one DT-helix HMM hit in 166 genomes, each of which was manually examined and evaluated using sequence support, position in the distal-tail neighborhood, preservation of the morphogenetic synteny, presence of the C-terminal helical extension, predicted fold, and association with a candidate receptor-binding gene (Table S2). This process retained 160 as confirmed and classified the remaining six as one borderline and five rejected candidates. Five rejected candidates were weak or compositionally spurious matches with incompatible lengths, folds, or genomic neighborhoods. The remaining borderline protein contained an atypical dimeric DT-core fold followed by a long C-terminal region with low confidence in the predicted structure (Fig. S2).

The set of 160 validated DT-helix proteins were clustered using

MMseqs2 (minimum sequence identity = 0.2, minimum alignment coverage = 0.2, clustering mode = “normal- > slow_sensitive”) into six clusters. Information about the cluster assignment is available at [https://github.com/tlensk/Sal3T, in Clusters]. Four of the six clusters contained 100 (A), 45 (B), 8 (C), and 5 (D) sequences respectively. The remaining two clusters (E and F) were singletons containing Vibrio phage Seahorse (NC_070772.1) and Acinetobacter phage vB_AbaS_TRS1 (NC_031098.1) respectively. Then, for each non-singleton cluster, a separate seed alignment was created from a reduced set of representative sequences generated by MMseqs2 (minimum sequence identity = 0.8, minimum alignment coverage = 0.8, clustering mode = “normal- > slow_sensitive”). For two singleton DT-helix clusters, the corresponding HMMs were built using one sequence. The 22 representative sequences employed for constructing the seed alignments from the final set of 6 HMMs were aligned using MUSCLE to construct a sequence tree using FastTree 2.2 (Price et al., 2010) (Fig. 3). The tree was rooted using lambda DT protein as an outgroup and visualized using iTOL v6 (Letunic and Bork, 2024). The six representative proteins employed for creating the HMM for the initial HMM screen were clustered as follows: (A) Jersey, FSL SP-031, K1H, and K1ind1; (B) Sf11; and (D) 9NA.

Then, the proteins in all 5167 genomes were screened again using this set of six DT-helix HMMs. The second run yielded 249 protein candidates from 245 genomes (most genomes had one protein candidate, and 4 genomes had 2 proteins with at least one HMM hit). Evidence-based evaluation according to the criteria in Section 2.6 of these genomes resulted in separating them in three categories (Table S3): confirmed (165), borderline (12), and rejected (68). The confirmed cases included 160 genomes with the curated DT-helix proteins from the initial HMM screen and 5 additional genomes with curated DT-helix proteins detected during the second HMM run. The borderline cases included: 1 borderline case from the initial HMM run and 11 new cases. The rejected cases consisted of 60 spurious hits and 8 DT proteins that did not have the extension. Notably, the rejected cases did not include any of the spurious hits reported as the result of the initial HMM run. The 165 curated DT-helix proteins were clustered again by MMseqs2 using the same parameters (minimum sequence identity = 0.2, minimum alignment coverage = 0.2, clustering mode = “normal- > slow_sensitive”) that were used for discovering the six clusters. Information about the cluster assignment for 165 curated DT-helix sequences is accessible at [https://github.com/tlensk/Sal3T, in Clusters].

Notably, one of the five additional positives identified only during the subgroup-HMM rescreen was the marine Pseudoalteromonas siphophage TW1. An independent cryo-EM study of TW1 had previously resolved six trimeric tailspikes attached to the distal tail complex and identified the corresponding gp15 protein as a Dit homolog related to lambda gpM and T5 pb9, with the tailspike gp19 proposed to attach through this component (Wang et al., 2018). Thus, recovery of TW1 by the refined HMM screen provides independent structural support for the expanded sequence-profile approach.

3.6. Structure-aware foldseek screening supports the curated DT-helical-extension set but reveals no additional distant homologues

The ProstT5/Foldseek screen of the local database containing 541,986 proteins yielded 458 protein candidates in 449 phage genomes across searches performed with 22 representatives of the six DT-helix sequence clusters (Table S4). The structure-aware search recovered proteins belonging to the established DT-helical-extension set, including the five additional DT-helix proteins identified during the refined six-HMM rescreen. In contrast, none of the 60 spurious HMM-hit proteins retained by the permissive rescreen was recovered by Foldseek. Focused follow-up searches also recovered three known DT-helix proteins that were not reported in the initial Foldseek screen, including the TRS1 DT protein and corrected full-length DT proteins from the closely related

Klebsiella phages KP591P1 (NC_071147.1) and KP591P3 (NC_071148.1).

The remaining Foldseek candidates with permissible protein length, genome context, and tail morphology were manually examined and did not provide convincing evidence for an additional highly divergent DT-helical-extension family outside the sequence groups already identified. Thus, within the 5167-genome RefSeq dataset, the structure-aware search did not reveal additional distant homologues that would expand the curated DT-helical-extension class. This result provides complementary support for the specificity and approximate completeness of the profile-based analysis within the dataset examined. Because the Foldseek database was generated from ProST5-predicted 3Di representations derived from amino-acid sequence, however, this analysis is not fully sequence-independent, and the absence of additional hits should not be interpreted as evidence that more divergent structural analogues do not exist.

3.7. DT sequence groups are associated with conserved tailspike attachment regions

To examine whether diversification of DT proteins is associated with corresponding sequence patterns in their cognate tailspikes, I compared the 165 DT proteins and associated tailspikes according to the six DT sequence clusters identified by MMseqs2. The clusters contained 100 (A), 48 (B), 9 (C), 5 (D), 1 (E), and 2 (F) proteins. The four principal multi-member DT clusters (A–D) showed pronounced within-cluster sequence conservation. Median full-length pairwise DT identities were 88.6%, 83.3%, 59.1%, and 77.8% for clusters A–D, respectively, whereas median identities between these clusters were only 22.2–27.1%. Cluster E was represented by a single protein, whereas the two members of cluster F shared only 22.7% full-length identity and therefore formed a substantially more divergent pair (Fig. S3a).

Next, I examined whether the tailspikes associated with these independently defined DT groups showed corresponding sequence relationships. Because the experimentally characterized Sf11 gp13 segment interacting with the DT lies within the proximal portion of the tailspike, pairwise sequence similarities were analyzed separately for the N-terminal 60 residues and for the remaining downstream sequence. For tailspikes associated with DT clusters A–D, median pairwise identities within the N-terminal region were 93.3%, 60.0%, 61.7%, and 75.8%, respectively (Fig. S3b). These values were substantially greater than those observed for the corresponding downstream regions, which had median identities of 46.7%, 28.4%, 31.3%, and 27.5%, respectively (Fig. S3c). Moreover, median N-terminal identities between different DT clusters were substantially lower, ranging from approximately 17–37%. Thus, the independently defined DT groups were associated with distinct sequence signatures in the proximal regions of their cognate tailspikes.

The contrast between proximal and downstream tailspike conservation was also evident at the level of individual sequence pairs. N-terminal identity exceeded downstream identity in 92.1%, 90.3%, 91.7%, and 100% of within-group comparisons for groups A–D, respectively. Cluster A showed particularly strong conservation of the proximal tailspike region despite broad variation in downstream sequence similarity, consistent with multiple divergent downstream tailspike subfamilies retaining a common proximal sequence signature. The two small DT clusters are interpreted separately. Cluster E contained only one DT–tailspike pair and therefore could not be evaluated for within-cluster conservation. The two cluster-F tailspikes shared only 16.4% identity within their N-terminal 60 residues and therefore did not show the strong within-cluster proximal conservation observed for groups A–D. Consequently, the association between DT-group identity and proximal tailspike conservation is strongly supported by the four principal multi-member clusters.

Together, these results reveal a correspondence between DT sequence diversification and sequence conservation in the proximal attachment region of the associated tailspike. Pairwise identity patterns

for the DT proteins, tailspike N-terminal 60 residues, and remaining downstream tailspike regions are shown in Fig. S3. The strong conservation of cluster-specific proximal tailspike signatures, accompanied by substantially greater diversification of downstream tailspike sequences, is consistent with sequence constraint at the morphogenetic attachment boundary between the DT scaffold and its cognate receptor-binding appendage. This organization could permit diversification of downstream tailspike regions involved in host interactions while retaining compatibility with the corresponding tail-tip assembly. The observed correspondence is consistent with coordinated diversification of the DT and proximal tailspike regions but does not by itself establish residue-level coevolution.

3.8. The architecture occurs across phages of Gram-negative hosts and encompasses multiple DT variants

Among the 165 curated phages, all phages with assigned hosts were associated with Gram-negative bacteria; the dataset included 18 named host genera and one record with an unknown host from a bat sample (Fig. 4a). Available morphology annotations and tail-gene organization were consistent with long, non-contractile tails. The positives were not restricted to a single currently recognized genus or subfamily of phages (Fig. 4b), supporting interpretation of the DT-helix system as a morphogenetic class rather than a formal taxonomic group.

Analysis of gene order across the 165 DT-helical-extension phages revealed a highly conserved tail-tip organization. All 165 genomes retained **TM, DT, TH, and CF in the same relative order**. Also, 161 of 165 (97.6%) contained **TNLP CDS** between TH and CF. Four genomes lacked an annotated CDS for TNLP which was rescued by manual curation. The common arrangement, **TM > DT > TH > TNLP > CF > tailspike**, was observed in 130/165 genomes (78.8%). Most deviations from this pattern reflected the presence of one or more intervening unlabeled ORFs rather than rearrangement of the conserved components. Such ORFs occurred in 35/165 genomes (21.2%) and were most frequently located between TM and DT (18 genomes), with less frequent insertions between DT and TH, TH and TNLP, TNLP and CF, or CF and the spike.

A tailspike CDS was present in 162/165 genomes (98.2%). In the remaining three genomes, the tailspike was close to the genome sequence ends and CDS was not annotated. These CDS were rescued fully or partially by manual curation. In 160 genomes, the spike was positioned downstream of CF in the same orientation as the tail-tip module, whereas five genomes had a tailspike encoded on the opposite strand. A single exceptional genome showed a different local arrangement in which the spike-like protein also occurred between DT and TH. Thus, despite occasional accessory ORFs, misannotation of TNLP, additional genes, or changes in spike orientation, the **TM–DT–TH–TNLP–CF** backbone was remarkably stable across the DT-helical-extension phages, supporting its interpretation as a conserved morphogenetic module.

3.9. PhageTailDB provides accession-level evidence for the curated classifications

To make the analysis accessible and auditable, I developed PhageTailDB, which records the 165 DT-helix phages at accession level. For each phage, the resource provides genome information, host and taxonomic information where available, as well as supporting details about the DT and associated tailspike or receptor-binding proteins. PhageTailDB therefore serves both as a catalogue of the identified architectures and as a starting point for selecting representative systems for molecular and synthetic studies.

4. Discussion

This study began with an annotation-level inconsistency and

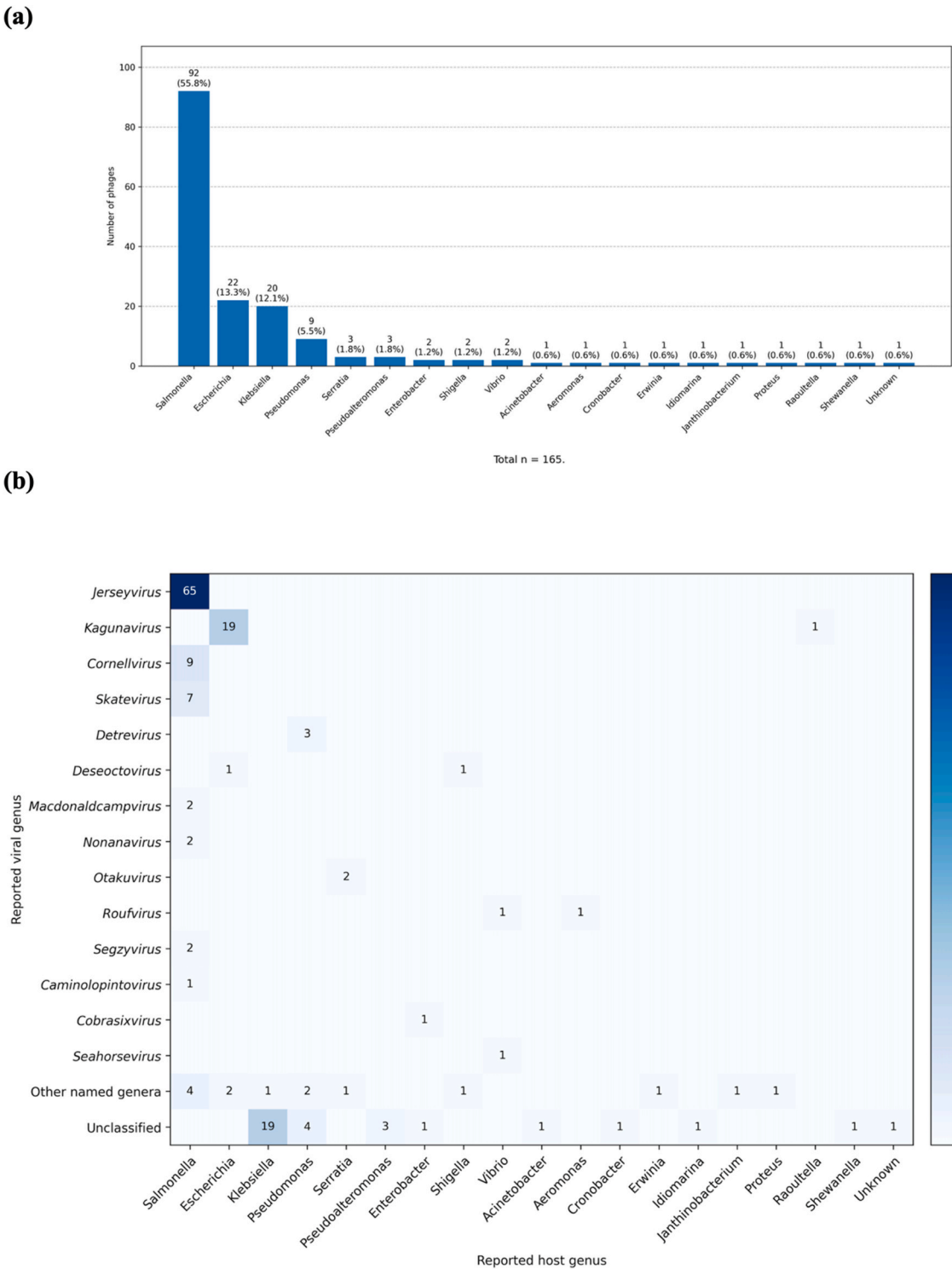


Fig. 4. Distribution of curated DT-helix phages: (a) by reported host genus; (b) by reported viral genus and host genus (see Table S6 for detail).

developed into the identification of a broadly distributed tail-tip architecture. The 9NA and Jersey genomes contained similarly positioned distal-tail genes despite different published labels. Comparison with lambda supplied a morphogenetic reference for assigning DT, TH, and TNLP roles, while the experimentally resolved Sf11 structure connected the C-terminal region of a related DT protein to tailspike attachment. Profile-based screening expanded this observation from a small set of paradigm phages to 165 curated architectures. Importantly, the independently defined DT sequence clusters were associated with

corresponding sequence signatures in the proximal regions of their cognate tailspikes, whereas downstream tailspike sequences were substantially more diverse. The proposed architecture is therefore supported by convergent evidence from DT sequence conservation, coordinated DT–tailspike sequence relationships, recurrent genomic context, the experimentally characterized Sf11 attachment interface, and complementary structural analyses.

The principal biological implication is that receptor-binding diversification appears to occur across a constrained morphogenetic

boundary rather than through unrestricted variation of the entire tail-tip assembly. The independently defined DT clusters were associated with distinct proximal tailspike sequence signatures, whereas sequence conservation declined substantially in the downstream portions of the tailspikes. This regional contrast suggests that evolutionary constraint is concentrated at the connection between the DT scaffold and its cognate receptor-binding appendage, while downstream tailspike regions can diversify more extensively. The experimentally resolved Sf11 complex provides a structural precedent for this interpretation, in which the proximal tailspike region attaches to the C-terminal portion of the DT protein. Together, these observations provide a plausible mechanism by which phages can preserve tail-assembly compatibility while diversifying receptor-facing functions.

The architecture is best treated as a morphogenetic and functional class, not necessarily as a formal taxonomic unit. Its distribution across phages of different Gram-negative hosts and multiple taxonomic lineages indicates that virion architecture can reveal relationships not captured by whole-genome taxonomy. This is particularly important in phage genomics, where rapid sequence divergence and inconsistent naming can hide conserved assembly logic. The study therefore supports an annotation strategy in which gene order, structural context, and experimentally characterized reference systems are considered together rather than treating each open reading frame as an isolated prediction problem. For synthetic phage engineering, the DT-helix identifies candidate boundaries between a conserved scaffold and a variable receptor-binding component. Current tail-fiber and tailspike engineering can modify host range, but successful replacements remain strongly context dependent. The present analysis does not show that all DT-helix-associated tailspikes can be exchanged, nor does it define universal compatibility rules. It does, however, reduce the search space by identifying natural systems in which conserved DT sequence architectures are coupled to corresponding proximal tailspike signatures while substantially greater diversification occurs downstream. These natural combinations provide candidate compatibility groups for testing whether receptor-binding modules can be exchanged within or between related DT–tailspike architectures.

TW1 provides an illustrative example of this modularity: although it is a siphophage, its six trimeric tailspikes are structurally related to the P22/Sf6/HK620-type beta-solenoid spikes of podophages, consistent with evolutionary exchange of receptor-binding modules across otherwise distinct tail architectures (Wang et al., 2018). The cluster-associated conservation of proximal tailspike sequences observed here provides a possible sequence-level mechanism for maintaining compatibility between such variable receptor-binding modules and the conserved distal-tail scaffold.

The relevance to phage therapy is therefore prospective rather than a claim of immediate therapeutic suitability. Host specificity is advantageous because it can limit off-target effects, but it complicates treatment when pathogenic isolates vary in receptor availability. DT-helix-containing phages may offer useful chassis or receptor-binding modules for future customization. Nevertheless, each candidate must still be evaluated for lytic lifestyle, absence of lysogeny, toxin and antimicrobial-resistance genes, transduction risk, propagation stability, host range, and performance in relevant infection models. The architecture is an engineering-relevant feature, not a substitute for these safety and efficacy criteria.

Several limitations define the current evidence. The interaction is experimentally established for Sf11 but computationally inferred for most members of the group. AF predictions can support an interface hypothesis but do not demonstrate binding, assembly, or productive infection. The discovery-stage HMM was constructed from six representatives, and the refined screen subsequently employed six group-specific profiles derived from the expanded sequence set. Although this substantially broadened sequence coverage, both profile-based and ProstT5/Foldseek searches remain biased toward sequence space represented in the current curated collection, and highly divergent

architectures may remain undetected. Host and taxonomic metadata in public records are incomplete, and the screen was restricted to complete RefSeq genomes, excluding many uncultivated and metagenome-derived phages. Finally, tailspike attachment is only one determinant of host range; adsorption, DNA delivery, intracellular defenses, and replication compatibility all influence productive infection.

The correspondence between DT groups and proximal tailspike sequence signatures does not establish direct residue-level coevolution or universal interchangeability; rather, it identifies a recurrent association. Future work should therefore test representative interfaces by cryo-EM, biochemical interaction assays, targeted mutagenesis, and engineered tailspike replacements. Computationally, the profile should be expanded iteratively, benchmarked by leave-one-lineage-out testing, and applied to broader genomic and metagenomic collections. Integrating receptor annotation and experimental host-range data into PhageTailDB may eventually permit evaluation of whether particular DT and tailspike features predict compatibility or receptor usage. The immediate contribution of this study is the definition of a testable architecture and a curated set of systems in which that hypothesis can be investigated.

5. Conclusion

Morphogenetic profiling identified a conserved C-terminal helical extension of distal tail proteins associated with tailspike attachment in 165 curated phage genomes. The architecture was first recognized in *Salmonella* phages, connected to experimental structural evidence from Sf11, and extended across phages of Gram-negative hosts through profile-HMM screening, genomic-context analysis, and structure prediction.

The DT-helix defines a candidate interface between conserved tail assembly and variable receptor binding. Although experimental exchangeability remains to be established, the architecture provides a rigorous framework for selecting phage scaffolds and receptor-binding modules for molecular characterization and rational engineering. PhageTailDB makes the underlying evidence available as a resource for phage biology, synthetic design, and future therapeutic development.

Declaration of Generative artificial intelligence

During the preparation of the manuscript the author used ChatGPT 5.5 as an assistive tool for proofreading and linguistic refinement. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Declaration of competing interests

The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.virol.2026.111060>.

Data availability

The phage information and supporting detail for 165 “DT-helix” phages are available in PhageTailDB v8.2 at [<https://tlensk.github.io/phagetaildb-dev/>; DOI: 10.5281/zenodo.21921997]. The HMM seed alignments, profiles, and raw search results are deposited at [<https://github.com/tlensk/Sal3T/>].

References

- Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Baillard, A.J., Bambrick, J., Bodenstein, S.W., 2024. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* 630 (8016), 493–500. <https://doi.org/10.1038/s41586-024-07487-w>.
- Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215 (3), 403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2).
- Altschul, S.F., Madden, T.L., Schäffer, A.A., Zhang, J., Zhang, Z., Miller, W., Lipman, D.J., 1997. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res.* 25 (17), 3389–3402. <https://doi.org/10.1093/nar/25.17.3389>.
- Anany, H., Switt, A.I.M., De Lappe, N., Ackermann, H.-W., Reynolds, D.M., Kropinski, A. M., Wiedmann, M., Griffiths, M.W., Tremblay, D., Moineau, S., 2015. A proposed new bacteriophage subfamily: “*jerseyvirinae*.”. *Arch. Virol.* 160 (4), 1021–1033. <https://doi.org/10.1007/s00705-015-2344-z>.
- Ando, H., Lemire, S., Pires, D.P., Lu, T.K., 2015. Engineering modular viral scaffolds for targeted bacterial population editing. *Cell Syst.* 1 (3), 187–196. <https://doi.org/10.1016/j.cels.2015.08.013>.
- Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., Madden, T.L., 2009. BLAST+: architecture and applications. *BMC Bioinf.* 10, 421. <https://doi.org/10.1186/1471-2105-10-421>.
- de Jonge, P.A., Nobrega, F.L., Brouns, S.J.J., Dutilh, B.E., 2019. Molecular and evolutionary determinants of bacteriophage host range. *Trends Microbiol.* 27 (1), 51–63. <https://doi.org/10.1016/j.tim.2018.08.006>.
- Dedrick, R.M., Guerrero-Bustamante, C.A., Garlena, R.A., Russell, D.A., Ford, K., Harris, K., Gilmour, K.C., Sothill, J., Jacobs-Sera, D., Schooley, R.T., Hatfull, G.F., Spencer, H., 2019. Engineered bacteriophages for treatment of a patient with a disseminated drug-resistant *Mycobacterium abscessus* infection. *Nat. Med.* 25 (5), 730–733. <https://doi.org/10.1038/s41591-019-0437-z>.
- Dunbrack Jr., R.L., 2025. ResipSAEloquent: what's wrong with AlphaFold's ipTM score and how to fix it. *bioRxiv*. <https://doi.org/10.1101/2025.02.10.637595>.
- Dunne, M., Rupf, B., Tala, M., Qabrat, X., Ernst, P., Shen, Y., Sumrall, E., Heeb, L., Plückthun, A., Loessner, M.J., Kilcher, S., 2019. Reprogramming bacteriophage host range through structure-guided design of chimeric receptor binding proteins. *Cell Rep.* 29 (5), 1336–1350.e4. <https://doi.org/10.1016/j.celrep.2019.09.062>.
- Eddy, S.R., 2011. Accelerated profile HMM searches. *PLoS Comput. Biol.* 7 (10), e1002195. <https://doi.org/10.1371/journal.pcbi.1002195>.
- Hardies, S.C., Park, J., Cho, B.C., Hwang, C.Y., 2026. Alishewanella phage LSH1 from the sea surface microlayer provides a novel minimalist view of the siphoviral hub structure. *Comput. Struct. Biotechnol. J.* 35 (1), 131. <https://doi.org/10.34133/csbj.0131>.
- Honorato, R.V., Trellet, M.E., Jiménez-García, B., Schaarschmidt, J.J., Giuliani, M., Reys, V., Koukos, P.I., Rodrigues, J.P., Karaca, E., van Zundert, G.C., Roel-Touris, J., 2024. The HADDOCK2.4 web server for integrative modeling of biomolecular complexes. *Nat. Protoc.* 19 (11), 3219–3241. <https://doi.org/10.1038/s41596-024-01011-0>.
- Jault, P., Leclerc, T., Jennes, S., Pirnay, J.-P., Que, Y.-A., Resch, G., Rousseau, A.F., Ravat, F., Carsin, H., Le Floch, R., Schaal, J.-V., Soler, C., Fevre, C., Arnaud, I., Bretaudeau, L., Gabard, J., CIFEPhAGE Study Group, 2019. Efficacy and tolerability of a cocktail of bacteriophages to treat burn wounds infected by *Pseudomonas aeruginosa* (PhagoBurn): a randomised, controlled, double-blind phase 1/2 trial. *Lancet Infect. Dis.* 19 (1), 35–45. [https://doi.org/10.1016/S1473-3099\(18\)30482-1](https://doi.org/10.1016/S1473-3099(18)30482-1).
- Kortright, K.E., Chan, B.K., Koff, J.L., Turner, P.E., 2019. Phage therapy: a renewed approach to combat antibiotic-resistant bacteria. *Cell Host Microbe* 25 (2), 219–232. <https://doi.org/10.1016/j.chom.2019.01.014>.
- Kutateladze, M., Adamia, R., 2010. Bacteriophages as potential new therapeutics to replace or supplement antibiotics. *Trends Biotechnol.* 28 (12), 591–595. <https://doi.org/10.1016/j.tibtech.2010.08.001>.
- Leitner, L., Ujmajuridze, A., Chanishvili, N., Goderdzishvili, M., Chkonia, I., Rigvava, S., Chkhotua, A., Changashvili, G., McCallin, S., Schneider, M.P., Sartori, A., Mehnert, U., Bachmann, L.M., Kessler, T.M., 2021. Intravesical bacteriophages for treating urinary tract infections in patients undergoing transurethral resection of the prostate: a randomised, placebo-controlled, double-blind clinical trial. *Lancet Infect. Dis.* 21 (3), 427–436. [https://doi.org/10.1016/S1473-3099\(20\)30330-3](https://doi.org/10.1016/S1473-3099(20)30330-3).
- Letunic, I., Bork, P., 2024. Interactive tree of life (iTOL) v6: recent updates to the phylogenetic tree display and annotation tool. *Nucleic Acids Res.* 52 (W1), W78–W82. <https://doi.org/10.1093/nar/gkae268>.
- Meng, E.C., Goddard, T.D., Pettersen, E.F., Couch, G.S., Pearson, Z.J., Morris, J.H., Ferrin, T.E., 2023. UCSF ChimeraX: tools for structure building and analysis. *Protein Sci.* 32 (11), e4792. <https://doi.org/10.1002/pro.4792>.
- Nobrega, F.L., Costa, A.R., Kluskens, L.D., Azeredo, J., 2015. Revisiting phage therapy: new applications for old resources. *Trends Microbiol.* 23 (4), 185–191. <https://doi.org/10.1016/j.tim.2015.01.006>.
- Price, M.N., Dehal, P.S., Arkin, A.P., 2010. FastTree 2—approximately maximum-likelihood trees for large alignments. *PLoS One* 5 (3), e9490. <https://doi.org/10.1371/journal.pone.0009490>.
- Schooley, R.T., Biswas, B., Gill, J.J., Hernandez-Morales, A., Lancaster, J., Lessor, L., Barr, J.J., Reed, S.L., Rohwer, F., Benler, S., Segall, A.M., Taplit, R., Smith, D.M., Kerr, K., Kumaraswamy, M., Nizet, V., Lin, L., McCauley, M.D., Strathdee, S.A., Benson, C.A., 2017. Development and use of personalized bacteriophage-based therapeutic cocktails to treat a patient with a disseminated resistant *Acinetobacter baumannii* infection. *Antimicrob. Agents Chemother.* 61 (10). <https://doi.org/10.1128/AAC.00954-17>.
- Stanaway, J.D., Parisi, A., Sarkar, K., Blacker, B.F., Reiner, R.C., Hay, S.I., Nixon, M.R., Dolecek, C., James, S.L., Mokdad, A.H., GBD 2017 Non-Typhoidal Salmonella Invasive Disease Collaborators, 2019. The global burden of non-typhoidal Salmonella invasive disease: a systematic analysis for the global burden of disease study 2017. *Lancet Infect. Dis.* 19 (12), 1312–1324. [https://doi.org/10.1016/S1473-3099\(19\)30418-9](https://doi.org/10.1016/S1473-3099(19)30418-9).
- Steinberger, M., Söding, J., 2017. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.* 35, 1026–1028. <https://doi.org/10.1038/nbt.3988>.
- Subramanian, S., Dover, J.A., Parent, K.N., 2025. The Shigella siphophage Sf11 tail structure and host attachment mechanism. *J. Virol.* 99 (12). <https://doi.org/10.1128/jvi.01367-25>.
- van Kempen, M., Kim, S.S., Tumescheit, C., Mirdita, M., Lee, J., Gilchrist, C.L., Söding, J., Steinberger, M., 2024. Fast and accurate protein structure search with foldseek. *Nat. Biotechnol.* 42 (2), 243–246. <https://doi.org/10.1038/s41587-023-01773-0>.
- Wang, C., Duan, J., Gu, Z., Ge, X., Zeng, J., Wang, J., 2024. Architecture of the bacteriophage lambda tail. *Structure* 32 (1), 35–46. <https://doi.org/10.1016/j.str.2023.10.006>.
- Wang, Z., Hardies, S.C., Fokine, A., Klose, T., Jiang, W., Cho, B.C., Rossmann, M.G., 2018. Structure of the marine siphovirus TW1: evolution of capsid-stabilizing proteins and tail spikes. *Structure* 26 (2), 238–248. <https://doi.org/10.1016/j.str.2017.12.001>.
- Waterhouse, A.M., Procter, J.B., Martin, D.M., Clamp, M., Barton, G.J., 2009. Jalview version 2—a multiple sequence alignment editor and analysis workbench. *Bioinformatics* 25 (9), 1189–1191. <https://doi.org/10.1093/bioinformatics/btp033>.
- World Health Organization, 2024. WHO bacterial priority pathogens list, 2024: bacterial pathogens of public health importance to guide research, development and strategies to prevent and control antimicrobial resistance. World Health Organization.
- Yehl, K., Lemire, S., Yang, A.C., Ando, H., Mimeo, M., Torres, M.T., de la Fuente-Núñez, C., Lu, T.K., 2019. Engineering phage host range and suppressing bacterial resistance through phage tail fiber mutagenesis. *Cell* 179 (2), 459–469.e9. <https://doi.org/10.1016/j.cell.2019.09.015>.
- Zeng, C., Gilcrease, E.B., Hendrix, R.W., Xie, Y., Jalfon, M.J., Gill, J.J., Casjens, S.R., 2019. DNA packaging and genomics of the Salmonella 9NA-like phages. *J. Virol.* 93 (22). <https://doi.org/10.1128/JVI.00848-19>.
- Zielezinski, A., Gudyś, A., Barylski, J., Siminski, K., Rozwalak, P., Dutilh, B.E., Deorowicz, S., 2025. Ultrafast and accurate sequence alignment and clustering of viral genomes. *Nat. Methods* 22 (6), 1191–1194. <https://doi.org/10.1038/s41592-025-02701-7>.